Exercise 3 11/10/2026

Question 1

Table: products.

SQL query

```
SELECT product_name,
       price,
       CASE
           WHEN price > 1000 THEN 'Expensive'
           WHEN price BETWEEN 100 AND 1000 THEN
           'Mid-ranged'
           WHEN price < 100 THEN 'Budget'
       END AS price_category
```

Table: Results

| product_name | price | price_category |
|---|---|---|
| Laptop | 1200.00 | Expensive |
| phone | 800.00 | Mid-range |
| Keyboard | 45.00 | Budget |
| Monitor | 300.00 | Mid-range |
| Mouse | 25.00 | Budget |



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Question 2

Table: Orders

SQL query

```
SELECT customer_name,
       amount,
       CASE
         WHEN amount >= 1000 THEN 'High Value'
         WHEN amount BETWEEN 500 AND 999.99
          THEN 'Medium Value'
         WHEN amount < 500 THEN 'Low Value'
       END AS order_value_category
FROM orders;
```

Table: Results

| customer_name | amount | order_value_category |
|---|---|---|
| Alice | 150.00 | Low Value |
| Bob | 560.00 | Medium Value |
| Charlie | 999.99 | Medium Value |
| Diana | 45.00 | Low Value |
| Ethan | 1200.00 | High Value |



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Question 3

Table: employees.

SQL query:

```
SELECT emp_name,
       department,
       salary,
       CASE
         WHEN department='IT' AND salary > 80000
         THEN 'Senior IT'
         WHEN department='HR' AND salary > 55000
         THEN 'Experienced HR'
         ELSE 'Staff'
       END AS position_level
FROM employees;
```

Table: Results

| emp_name | department | salary | position level |
|---|---|---|---|
| John | IT | 85 000 | Senor IT |
| Sara | HR | 60 000 | Experienced HR |
| Mark | IT | 75 000 | Staff |
| Lucy | Finance | 95 000 | Staff |
| Tom | HR | 55 000 | Staff |



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Question 4

Table: students

SQL Query:

```
SELECT student_name,
       score,
       CASE
         WHEN score >= 90 THEN 'A'
         WHEN score BETWEEN 80 AND 89
         THEN 'B'
         WHEN score BETWEEN 70 AND 79
         THEN 'C'
         WHEN score BETWEEN 60 AND 69
         THEN 'D'
         WHEN score < 60 THEN 'F'
       END AS grade
FROM students;
```

Table: Results

| student_name | score | grade |
|---|---|---|
| Anna | 92 | A |
| Ben | 76 | C |
| Carl | 59 | F |
| David | 83 | B |
| Ella | 68 | D |



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Question 5

Table: deliveries

SQL Query

```
SELECT delivery_id,
       delivery_time_minutes,
       CASE
         WHEN delivery_time_minutes <=30
         THEN 'Fast'
         WHEN delivery_time_minutes BETWEEN 31
         AND 60 THEN 'On Time'
         WHEN delivery_time_minutes > 60 THEN 'Late'
       END AS performance
FROM deliveries;
```

Table: Results

| delivery_id | delivery_time_minutes | performance |
|---|---|---|
| 1 | 45 | On Time |
| 2 | 80 | Late |
| 3 | 30 | Fast |
| 4 | 65 | Late |
| 5 | 100 | Late |



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Question 6

Table: tickets

SQL Query

```
SELECT issue_type,
       priority,
       CASE
       WHEN priority = 3 THEN 'High'
       WHEN priority = 2 THEN 'Medium'
       WHEN priority = 1 THEN 'Low'
       END AS priority_label
FROM tickets;
```

Table: Results

| issue_type | priority | priority_label |
|---|---|---|
| Login issue | 1 | Low |
| Server down | 3 | High |
| Slow system | 2 | Medium |
| Email error | 2 | Medium |
| Password Reset | 1 | Low |



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Question 7

Table : attendance

SQL Query

```
SELET student_id,
    Round (days_present x 100.0) / total_days, 2) AS
    CASE
      WHEN (days_present x 100.0) / total_days >= 90
      THEN 'Excellent'
      WHEN (days_present x 100.0) / total_days BETWEEN
      75 AND 89.99 THEN 'Good'
      WHEN (days_present x 100.0) / total_days < 75
      THEN 'Needs Improvement'
      END AS attendance_status
FROM attendance;
```

Table: Results

| student_id | attendance_percentage | attendance_status |
|---|---|---|
| 1 | 90.00 | Excellent |
| 2 | 60.00 | Needs Improvement |
| 3 | 96.00 | Excellent |
| 4 | 50.00 | Needs Improvement |
| 5 | 100.00 | Excellent |

Calculations

Student 1 : 45/50 = 90.00% = Excellent

Student 2: 30%



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Calculations

Student 1 => 45 ÷ 50 = 90.00% (Excellent)
Student 2 => 30 ÷ 50 = 60.00% (Needs Improvement)
Student 3 => 48 ÷ 50 = 96.00% (Excellent)
Student 4 => 25 ÷ 50 = 50.00% (Needs Improvement)
Student 5 => 50 ÷ 50 = ~~100%~~ 100.00% (Excellent)

Question 8

Table: product_inventory

SQL Query

```
SELECT product_id,
       stock_qty,
       CASE
         WHEN stock_qty = 0 THEN 'out of stock'
         WHEN stock_qty BETWEEN 1 AND 5
         THEN 'Low Stock'
         WHEN stock_qty > 5 THEN 'In Stock'
       END AS stock_status
FROM products_inventory;
```

Table: Results

| product id | stock_qty | stock_status |
|---|---|---|
| 1 | 5 | Low stock |
| 2 | 0 | Out of Stock |
| 3 | 25 | In stock |
| 4 | 10 | In stock |
| 5 | 3 | Low stock |



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Question 9

Table: classes

SQL Query

```
SELECT subject,
       enrolled_students,
    CASE
       WHEN enrolled_students >= 25 THEN 'Large'
       WHEN enrolled_students BETWEEN 10 AND 24
       THEN 'Medium'
       WHEN enrolled_students < 10 THEN 'Small'
    END AS class_size_category
FROM classes;
```

Table: Results

| subject | enrolled_students | class_size_category |
|---|---|---|
| Math | 30 | Large |
| English | 25 | Large |
| Science | 15 | Medium |
| Art | 5 | Small |
| History | 20 | Medium |



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Question 10

Table: payments

SQL Query

```
SELECT payment_id,
       payment_method,
       amount,
       CASE
         WHEN payment_method = 'cash' AND amount
         >= 200 THEN 'Eligible for Discount'
         ELSE 'Not Eligible'
       END AS discount_eligibility
FROM payments;
```

Table: Results

| payment_id | payment_method | amount | discount_eligibility |
|---|---|---|---|
| 1 | Card | 50.00 | Not Eligible |
| 2 | Cash | 200.00 | Eligible for Discount |
| 3 | Card | 150.00 | Not Eligible |
| 4 | Paypal | 75.00 | Not Eligible |
| 5 | Cash | 300.00 | Eligible for Discount |



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